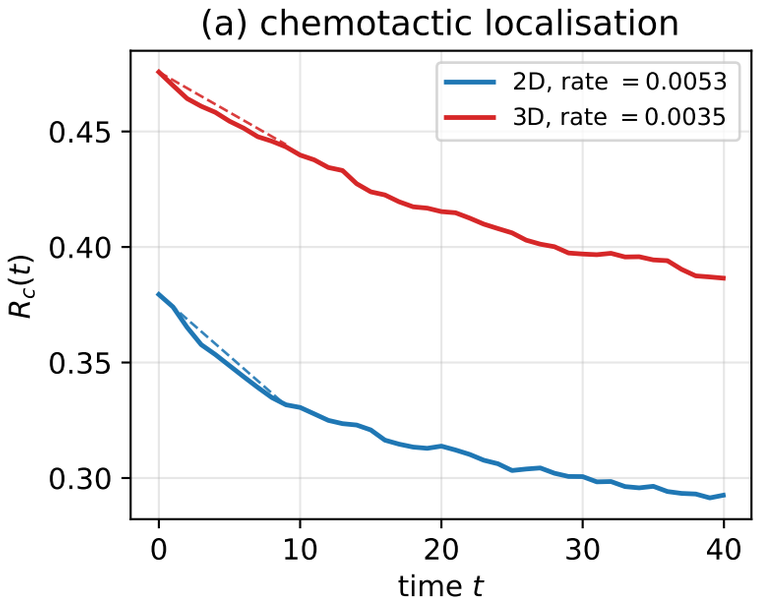
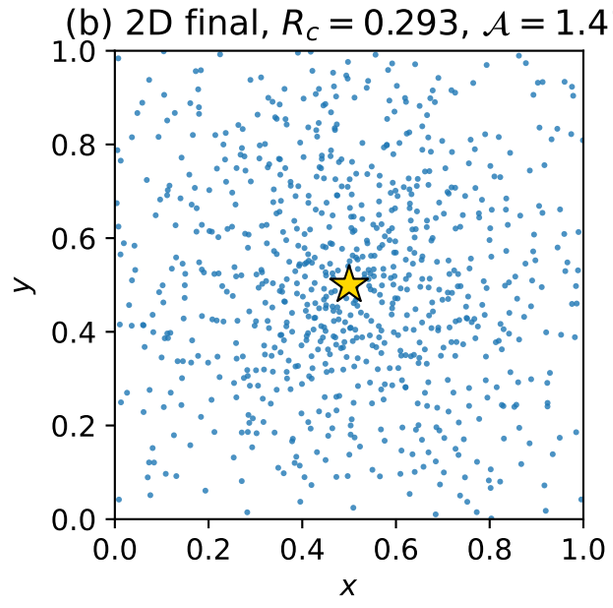
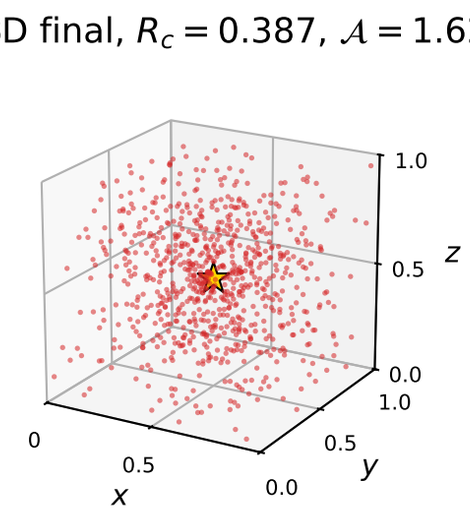
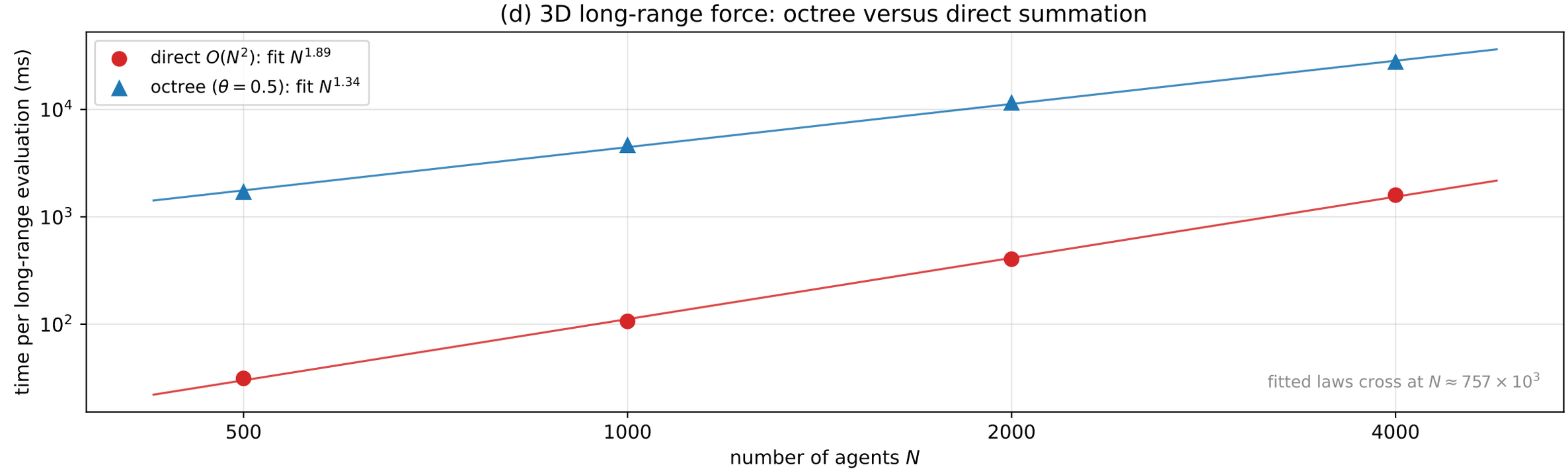

(a) chemotactic localisation
2D, rate = 0.0053
3D, rate = 0.0035
$R_c(t)$
time t


(b) 2D final, $R_c = 0.293$, $\mathcal{A} = 1.41$
x
y


(c) 3D final, $R_c = 0.387$, $\mathcal{A} = 1.62$
x
y
z


(d) 3D long-range force: octree versus direct summation
direct $O(N^2)$: fit $N^{1.89}$
octree ($\theta = 0.5$): fit $N^{1.34}$
time per long-range evaluation (ms)
number of agents N
fitted laws cross at $N \approx 757 \times 10^3$